VYNVORI / ENGINEERING PLAYBOOK

AI Model Benchmark Execution Blueprint

A coding-agent handoff for evaluating image generation, editing, print suitability and prepaid-credit economics.

MISSION

Implement and run the seven-step, reproducible benchmark for Vynvori. Return auditable results, ranked models, accepted sample assets, cost evidence, and a proposed credit price table for independent founder review.

This is a bounded research workstream. Do not implement the full e-commerce storefront, purchase production inventory, spend beyond the approved API cap, or silently declare model winners.

Execution at a glance

PHASE	REQUIRED EVIDENCE
01 Provider research	API access, model catalog, pricing, rights, capability matrix
02 Evaluation dataset	Four repeatable briefs, original or licensed references, expected output specs
03 Benchmark harness	Runnable TypeScript adapters, usage logging, guardrails, reproducible runs
04 Artwork evaluation	Blinded visual scoring, edit-fidelity tests, print-readiness checks
05 True cost accounting	Invoice-aligned actual spending and cost per accepted image/design
06 Credit economics	3–4× sensitivity, packages, standard/premium renders, shirt redemption
07 Product integration plan	ModelRouter contract, rollout strategy, risks and roadmap for later build

Scope: Gaming / ARPG creative briefs; black oversized/drop-shoulder shirt first. Standard and premium customer-selectable final render. US/EU/Australia commercial ambition.

Handoff condition: Deliver the full result-document folder described in Appendix A, then return to the founders for model selection.

Operating contract for the coding agent

COPY THIS INTO THE AGENT TASK

Objective: Build and execute a fair, multi-provider benchmark for Vynvori. Follow Steps 1–7. Maintain a verifiable research log; measure actual charges; keep prompts and reference sets comparable; produce the named reports and machine-readable result files. Never infer a winner from model reputation or list prices alone.

Execution control: Begin in dry-run mode. Require explicit operator confirmation for credentials, any paid run, and spend caps. Stop before exceeding the cumulative budget. Do not automate providers that prohibit it.

Final handoff: Produce the required report directory, a concise decision memo, and the full raw results (with secrets and personal data removed). Label untested, blocked, or estimated evidence explicitly.

Already-approved Vynvori decisions — do not revisit

AREA	LOCKED REQUIREMENT
Infrastructure	Next.js + Vercel-managed Workflow SDK 5 / AI SDK 7; GCP Cloud Run for heavy processing; Cloudflare R2 for assets.
Data	Neon PostgreSQL + Drizzle ORM / Drizzle Kit + node-postgres pooled connections.
Experience	Visual moodboard wizard + optional creative chat; bounded agent autonomy; guided workflow.
Generations	Three initial previews, two refinements, final render: standard or customer-selectable premium.
Commerce	Prepaid credits; minimum proposed top-up US$5; one limited promotional preview; 3–4× effective-cost target.
Garment	Black oversized/drop-shoulder T-shirt first; 2D editor with front/back/sleeve zones; no virtual try-on for MVP.
Rights / community	Gallery in MVP; public display separate from permissions to resell; no unauthorized gaming IP.
Incentive	Partial eligible design-spend redemption against the corresponding physical shirt; rate not yet decided.

Roles and approval gates

- Coding agent: research, implement, test, collect evidence, fill reports, identify uncertainty and stop at budget or compliance gates.
- Human operator: supply API credentials via environment variables, approve budget, resolve provider access, choose which reference images can be used, commission physical print tests.
- Founders + ChatGPT later: judge design samples blind, approve model tiers and pricing, resume business planning after receiving results.

Budget and scope safeguards

- Default run cap: US$0 in dry run. Suggested pilot US$30–50 total, only after explicit approval; track separate caps for provider and stage.
- Planned pilot: 32 concept images + 16 edit attempts + 8 final render attempts = 56 image operations. These are operations, not guaranteed usable designs.
- Limit concurrency and attempts; record every billable request including rejected, failed, and repeated jobs.

- Never commit API keys to source control, logs, PDFs, screenshots, or final handoff. Use synthetic users, lawful moodboards, and non-infringing artwork.

Step 1 / Verify providers and current APIs

Goal: establish a fact-checked, reproducible shortlist before any paid image request. Do not rely on provider names or prices from previous chats as immutable facts.

Actions

1. Inspect the actual project environment. Create a dedicated packages/ai-benchmark module; record runtime, package versions, and package-manager lockfile.
2. Check official live model catalogs, exact API model IDs, authentication, availability, commercial-use terms, output licenses, safety limits, region limitations and rate limits.
3. Cover at least Google Gemini / Nano Banana, OpenAI image API, fal / FLUX, Higgsfield API, and Ideogram. Treat Midjourney as a manual-only optional artistic reference unless official authorization for automation is documented.
4. For each relevant model, record generate vs edit capability, multi-image references, masking, output transparency, exact resolution/aspect ratios, watermark behavior, batch support, seed support, async execution and result URL expiration.
5. Collect per-operation billing rules: text/image inputs, output image tokens or pixels, quality settings, reference count effects, retries, upscale costs, minimum charges and refunds. Record the date, currency, and primary URL.
6. Rank on eligibility, not reputation; select 4 primary concept candidates and 2–4 edit/final candidates. If the same model is offered direct and via an aggregator, record both effective costs and routing terms.

Hard gates and exceptions

- No unofficial Midjourney automation or account sharing. See its official guidelines in the references.
- If exact pricing/API capability is not public or current, mark UNVERIFIED and ask the operator; do not fabricate estimates and claim they are quotes.
- Only choose models you can call with authorized commercial API access; manual benchmark items must remain outside automated rankings.

Required output / exit criteria

STEP 1 ARTIFACTS

docs/01-provider-matrix.md - human-readable comparisons, trade-offs, rejected options.

data/model-manifest.json - machine-readable provider/model/operation configuration and source URLs.

docs/01-cost-assumptions.md - dated list prices, billing interpretation, uncertainty and approved spend cap.

Exit: at least four eligible primary models, truthful current API IDs, credentials tested without paid requests where feasible, no missing pricing provenance.

Step 2 / Build the common evaluation dataset

Goal: ensure the comparison measures models, not changes in creative briefs, inappropriate source material, or selective cherry-picking.

Dataset specification

CASE ID	CORE CREATIVE TASK	WHAT IT TESTS
ARPG-01	Original dark-fantasy graphic with the generic phrase “I’m only mapping.”	Wearability; motif relevance; exact phrase composed in an editable text layer.
MIN-02	Minimal gothic/industrial geometry for a black streetwear T-shirt.	Composition and restraint, negative-space discipline, legibility.
CHAR-03	Original fictional ARPG-inspired mascot with consistent silhouette.	Character integrity and local-edit retention across revisions.
MOOD-04	Synthesize 3–5 licensed/original references into one psychedelic/fantasy motif.	Multi-image moodboard fidelity without copying any particular reference.

Actions

1. Create four JSON fixture briefs with stable IDs and version numbers; include user intent, palette, symbolism, exclusions, art direction, black-shirt context, target zone, and output format.
2. For each, assemble clearly licensed or internally created moodboard images; record source, license/right to use, checksum, image dimensions and usage constraints. Use no third-party game logos, identifiable copyrighted characters, or unlicensed photographs.
3. Create three prompt forms per case: (a) vendor-neutral core, (b) provider-adjusted prompt preserving identical meaning, and (c) targeted refinement instruction. Save both authored and sent prompt text.
4. Define common preview output target (e.g., square ~1024 px, or closest supported) and a comparable high-quality render target; record unavoidable native differences rather than silently changing requests.
5. Define two edit tests: controlled color/motif edit; and localized edit while preserving character/composition. Exact phrases should be handled by typography/compositing logic, not assumed error-free inside AI-generated raster art.
6. Use fixed generation seeds where supported, but do not claim determinism across providers. Plan two independent concept trials per model/brief; randomize display order for later blind review.

Required output / exit criteria

STEP 2 ARTIFACTS

fixtures/briefs/*.json, fixtures/references/ and fixtures/references/LICENSES.csv.

docs/02-dataset-and-protocol.md - complete fairness protocol, prompt templates, trial matrix, intended sample count.

Exit: all four cases reproducible without private user data; all sources licensed; reference files checksummed; prompt semantics consistent.

Step 3 / Implement and execute the benchmarking harness

Goal: produce a runnable TypeScript CLI with provider adapters, strict accounting, reliable storage and deterministic task IDs. Build the benchmark as a reusable module, not a storefront feature.

Implementation structure

```
packages/ai-benchmark/
  src/providers/{google,openai,fal,higgsfield,ideogram}.ts
  src/{model-router,runner,cost-meter,validators,report}.ts
  src/evaluators/{technical,comparison}.ts
  fixtures/{briefs,references}/
  config/{models,benchmark}.json
  scripts/{dry-run,run-pilot,export-results}.ts
  results/{runs,artifacts,report}/
  tests/{adapters,ledger,workflow}.test.ts
```

Actions

1. Use Vercel AI SDK 7 interfaces where the provider supports the required operation. Add a typed native REST adapter for endpoints requiring provider-specific generate/edit/async semantics; normalize outputs into one internal response schema.
2. Implement a capabilities registry and explicit model-operation mapping. Reject unsupported masking, transparency, reference counts or sizes before calling an API; never silently substitute providers or request qualities.
3. Introduce a CLI run manifest with dry-run default, authorized model list, per-provider cap, global cap, concurrency limit, timeout, retries, output dimensions and experiment version.
4. Each request gets a unique run ID and provider correlation ID. Store start/end times, prompt hash and redacted text, source image hashes, output metadata, SDK warnings, error type and charge evidence.
5. Persist image bytes immediately into R2 or a local durable artifact directory; record SHA-256 digest. Never trust temporary provider-hosted URLs as permanent artifacts.
6. Use bounded retries for transient failures; avoid blind resubmission of ambiguous paid requests. Implement job states queued/running/completed/failed/unknown and a manual reconcile path.
7. Implement export to CSV and JSONL. Include pricing-source version, actual billed amount when available, and estimated amount with an explicit evidence class if the invoice is pending.
8. Run automated tests with fake providers first; then do one authorized smoke test per selected vendor; finally run the staged pilot without exceeding the approved cap.

Common result record — minimum required fields

```
{
 "runId":"uuid", "caseId":"ARPG-01", "trial":1,
 "provider":"...", "modelId":"exact-id", "operation":"concept|edit|final",
 "briefVersion":"v1", "promptSha256":"...", "inputAssetHashes":[],
 "requested": {"size":"1024x1024", "quality":"..."},
 "output": {"assetPath":"...", "sha256":"...", "width":1024,
            "height":1024, "format":"png", "hasAlpha":true},
 "latencyMs":12000, "attempts":1, "status":"completed",
 "billing": {"amountUsd":0.05, "evidence":"invoice|usage|estimate"},
 "providerRequestId":"...", "warnings":[], "errorCode":null
}
```

Required output / exit criteria

STEP 3 ARTIFACTS

Runnable repository package with README commands, .env.example, locked dependencies, unit tests and dry-run evidence.

results/runs/*.jsonl, results/artifacts/, results/manifest.csv, docs/03-execution-log.md.

Exit: reproducible execution from a clean install, zero exposed credentials, all calls cost-tracked, manifest links every generated file to its inputs.

Step 4 / Evaluate creativity, edit fidelity and printability

Goal: select artwork that is not merely attractive on screen, but appropriate for a premium black T-shirt and reliably editable.

Blind human evaluation

1. Create an anonymous review set. Replace provider/model names with randomized sample IDs, retain the hidden mapping outside the scoring sheet, and use an identical display canvas and garment context.
2. Ask two independent reviewers (ideally founders) to score each candidate 1–5. Review without seeing model price, name or speed. Resolve major disagreements with a documented adjudication.
3. Use this scoring rubric: artistic originality 25%, moodboard fidelity 20%, localized edit/identity preservation 20%, print suitability 20%, cost per accepted output 10%, latency/reliability 5%.
4. Use an objective print gate in addition to weighted scores. A non-printable output cannot win the production category regardless of artistic score.
5. Select up to two winning samples per category for a physical print feasibility test on the actual intended black garment/printing method; record whether any color or edge corrections were required.

Print-readiness inspection checklist

- Artwork-only source: design generated independently of a shirt photograph; no garment seams, fake printing shadows, mockup watermark or background fragments embedded in production artwork.
- Format + alpha: inspect PNG/WebP/TIFF support, real transparency vs checkerboard imagery, haloing and fuzzy semi-transparent edges; maintain original plus production derivative.
- Scale: record native pixels, planned inches/mm, resulting effective PPI. A2 is 420 × 594 mm (approximately 4,961 × 7,016 pixels at 300 PPI) and is a target print area only where the supplier confirms it.
- Separations and color: assess black-fabric visibility, white underbase needs, fine gradients, line thickness, text edges, print method compatibility and washed/pressed appearance.
- Placement: use printable-zone geometry for front/back/sleeves. Do not claim a universal A2 print area on small sizes or sleeves.
- IP / originality: check for recognizable game logos or characters, close copying of references, disallowed text, and unlicensed reference elements.

Scoring boundaries

METRIC	METHOD	DECISION RULE
Human art score	2 blinded raters, 1–5 scale, all four cases	Report median, range, inter-rater gap, examples.
Edit fidelity	Preservation rubric + targeted-difference notes	Report unintended changes and user-request satisfaction.
Technical print checks	PPI, format, alpha, artifact inspection	PASS / CONDITIONAL / FAIL with exact reason.
Reliable delivery	Success count, retries, p50/p95 latency	Report all runs including timeouts and failures.
Effective cost	Accounted USD / accepted outputs	Separate by task + standard/premium tiers.

Required output / exit criteria

STEP 4 ARTIFACTS

docs/04-evaluation-rubric.md, results/report/blind-scores.csv, docs/04-print-readiness.md, results/report/contact-sheet.pdf.

Optional physical: photos of actual prints with printer, fabric, dimensions and transfer-method records, clearly separated from digital-only evaluation.

Exit: ranked models by operation, annotated pass/fail evidence, unresolved design flaws documented, no unblinded subjective scores.

Step 5 / Calculate true economics and select model tiers

Goal: measure the cost of a design the customer accepts, rather than treating a provider list price as the actual business expense.

Cost allocation and reconciliation

1. Export provider billing/usage records for the experiment period. Reconcile each API charge with run IDs where possible. Separate confirmed charges, provider-reported estimates and unreconciled charges.
2. Calculate for each operation: model input cost + output cost + any upscale/mask/reference premium + failed attempts + network/processing allocation + asset storage share.
3. Compute per-model pass rate and cost per accepted output (all associated costs divided by accepted count). Include cost of discarded/failed outputs in the numerator.
4. Model a full session using the approved bundle: three concepts, two refinements and one final render; calculate separately for standard and premium final tiers. Show typical vs heavy-retry users.
5. Run pricing sensitivity at 3.0×, 3.5× and 4.0× total effective allocated generation cost. Recheck payment-fee effect and compare against published competitor customer prices.
6. Identify the least expensive qualified model by task. If a more expensive model has better acceptance and lower cost per accepted design, document the reversal.
7. Flag uncertainty caused by low sample counts. Do not extrapolate a 56-operation pilot into population-level claims or guaranteed margins.

Mandatory formulas

ECONOMICS THAT MUST APPEAR IN THE REPORT

Cost per accepted output = (all charged API requests + retry/failure costs + allocated preprocessing/postprocessing) / number of accepted outputs.

Session cost = costs of 3 concepts + 2 refinements + chosen final render + creative-director/prompt costs + storage/processing allocation.

Retail list target = 4 × effective session cost; discount floor under consideration = 3 × cost, subject to payment fees, redemption, taxes and full-order unit economics.

Contribution after fee = net paid design revenue - generation expense - payment fees - redemption subsidy - variable support/processing.

Required output / exit criteria

STEP 5 ARTIFACTS

results/report/cost-reconciliation.csv, docs/05-unit-economics.md and results/report/model-leaderboard.csv.

Deliver base / conservative / high-retry scenarios and an explicit USD per accepted design by operation and complete session.

Exit: actual vs estimated costs distinguished; model tier recommendations supported by quality gates and cost evidence.

Step 6 / Propose a prepaid-credit economy

Goal: use empirical costs and competitive pricing to propose, not silently activate, the Vynvori credit model.

Pricing workplan

1. Reconfirm current consumer-facing prices and capabilities for Krea, OpenArt, Ideogram, Midjourney (manual competitor), AI shirt designers and custom print storefronts. Separate subscription effective cost from API cost and from physical shirt price.
2. Choose a candidate credit denomination only after the measured standard and premium session costs are known; compare integer usability, minimum top-up $5 and customer comprehension.
3. Price individual preview, edit, final-standard, final-premium and optional digital-download events with the target 4× effective cost and a tested 3× discount boundary.
4. Propose one guided bundle (3 concepts + 2 edits + standard final) and a premium upgrade using live measured costs. Show final price in credits and USD with exact top-up requirements.
5. Include one free first-preview incentive and calculate trial conversion break-even at different conversion rates; treat its cost as acquisition spend.
6. Model partial design-spend redemption against a later physical shirt as a separately recorded discount entitlement. Do not choose the percentage before full garment landed margins are measured.
7. Define refunds, errors, credits on cancellation, failed renders, unused balances, cross-border taxation uncertainty, chargebacks and creator download rights.

Illustrative calculation — NOT customer pricing

VARIABLE	EXAMPLE	RULE
Effective session cost	$0.40	Illustrative only; replace with experiment results.
4× list target	$1.60	Before tax, gateway fees and redemption.
3× promo floor	$1.20	Use only if post-fee and total economics permit.
Prepaid package	$5 minimum	Number of credits decided only after modeling.
T-shirt redemption	TBD	Must be capped by audited garment margin.

Required output / exit criteria

STEP 6 ARTIFACTS

docs/06-market-pricing.md - customer prices, product scope, checked dates, official URLs.

docs/06-credit-pricing-proposal.md + results/report/credit-simulator.csv - credits/USD, operation prices, standard/premium bundles, sensitivity.

Exit: all proposal values trace to evidence; no credit price or promotional discount silently set live; separate unpriced shirt-redemption assumptions.

Step 7 / Produce the integration-ready engineering plan

Goal: make the benchmark winner straightforward to add to Vynvori without building the entire storefront during this research subproject.

Actions

1. Design a versioned ModelRouter mapping operation + quality tier + capability requirements to the qualified provider/model. Add safe fallback routes and fail-closed behavior when no compliant model exists.
2. Define specialist contracts for Moodboard Interpreter, Creative Director, Prompt Curator, provider image tool and quality checks; keep the orchestrator bounded by workflow/credit authorization rather than prompt instructions alone.
3. Design Vercel Workflow SDK jobs for concept generation, refine and final render. Separate agent reasoning from financial writes; store provider requests, credits and approved design versions in Neon.
4. Write Neon/Drizzle schema recommendations for ModelCatalog, ModelQualification, BenchmarkRun, GenerationJob, DesignVersion, CreditWallet, CreditLedger and UsageCharge with idempotency keys.
5. Specify the R2 asset lifecycle: temporary references, signed access, versioning, approved print-art original, print derivative and purge policy. Ensure generated artwork never inherits unlicensed reference assets.
6. Write a migration-safe rollout plan: shadow evaluations, feature flags, per-tier routing, spend alerts, provider outage behavior, cost anomaly detection and telemetry dashboards.
7. Implement the provider adapters and router proof-of-concept in the benchmark module. Leave checkout, user gallery, full mockup editor, wallet charging and production deployment as follow-on business-plan implementation.
8. Declare a recommended standard pipeline, premium pipeline, alternatives, and explicit blockers (such as insufficient print quality or incompatible edit APIs).

Required output / exit criteria

STEP 7 ARTIFACTS

docs/07-production-integration.md, docs/07-model-router-contract.md and docs/07-rollout-checklist.md.

Code: typed adapters, capability registry, routing proof-of-concept, tests, configuration with pinned model versions.

Exit: a new coding agent can implement the production pipeline from the docs without guessing model capabilities, financial behavior or asset lifecycles.

Appendix A / Required final handoff package

The coding agent must return a single folder or ZIP preserving this structure. All report files must exist even when the corresponding test was blocked; in that case explain the blocker and mark values N/A.

```
vynvori-ai-benchmark-results/
  README.md                     # start here: executive summary + status
  docs/
    01-provider-matrix.md
    01-cost-assumptions.md
    02-dataset-and-protocol.md
    03-execution-log.md
    04-evaluation-rubric.md
    04-print-readiness.md
    05-unit-economics.md
    06-market-pricing.md
    06-credit-pricing-proposal.md
    07-production-integration.md
    07-model-router-contract.md
    07-rollout-checklist.md
    08-founder-decision-memo.md
  data/model-manifest.json
  fixtures/briefs/*.json
  fixtures/references/LICENSES.csv
  results/runs/*.jsonl
  results/artifacts/*
  results/report/{manifest,blind-scores,cost-reconciliation,
                model-leaderboard,credit-simulator}.csv
  results/report/contact-sheet.pdf
  code/                         # benchmark harness + tests
  environment.md                # package versions, command log, redacted env
  SOURCES.md                    # official URLs, access date, scope of claim
```

08-founder-decision-memo.md — required headings

1. Answer: Best qualified model for promotional preview, concepts, refinements, standard final, premium final; include runner-up and downgrade fallback.
2. Evidence: blind art score, print gate result, accepted-output count, actual USD cost, p50/p95 latency and verified API price source.
3. Pricing: recommended credit denomination, minimum $5 pack, operation table, complete bundle, 3×/4× sensitivity, free-preview expense.
4. Known risks: low sample size, ambiguous billing, provider restrictions, region limits, image artifacts, external rights and opaque acceptance scoring.
5. Decisions required from founders: final model tier selection, API budget increase, physical print test, digital-art rights, redemption cap and go/no-go for implementation.
6. Suggested next business-plan work: resume Vynvori six-phase roadmap; do not repeat this benchmark unless evidence is inconclusive or models materially change.

Minimum summary metrics

MEASURE	REQUIRED UNIT / RULE
Success / acceptance	n completed, n accepted, rejection reasons by model-operation.
Cost	Confirmed and estimated USD; cost/request; cost/accepted; full standard/premium session.
Reliability	Failure %, retry count, timeouts, p50/p95 latency, unverified usage.
Creative scores	Blind human scores; range and disagreement; edit preservation.
Print	Clear PASS/CONDITIONAL/FAIL; selected sample asset paths and defects.
Provider viability	Official API and commercial rights, limits, known policy constraints.
Economics	3× / 3.5× / 4×, payment fees, prepaid pack, trial cost, redemption sensitivity.

Appendix B / Acceptance criteria and stop conditions

A deliverable counts as successful only if

- At least four currently supported primary candidates were fairly compared, or blockers and replacement rules are fully documented.
- Every scored image is tied to an exact model ID, prompt/fixture revision, run record and durable file.
- Both generation and editing are evaluated; final-render suitability is not inferred from previews alone.
- Print-quality failures cannot be masked by subjective beauty scores.
- Money spent is reconciled to provider usage or clearly marked provisional; cost per accepted design is calculated.
- Pricing proposals include 3×/4× sensitivities and no silently invented credit denomination.
- A founder can reproduce at least a smoke test from a clean checkout using README commands and credentials they supply.

Stop and ask the operator if

- The approved total spend would be exceeded, an account lacks access, or provider pricing is missing/contradictory.
- A model requires an unofficial or prohibited API path, or the requested references cannot be used lawfully.
- A paid request has an unknown completion/billing state and rerun may double-charge.
- Two high-quality print samples cannot meet garment production specifications, making final model selection premature.
- A new framework version breaks planned integrations; isolate and document the breaking contract rather than silently changing the locked platform.

Appendix C / Handoff to the main six-phase venture plan

This benchmark is a discrete technical research assignment. After review, return to the six-phase Vynvori planning track rather than expanding the research indefinitely.

MAIN PHASE	STATUS / HOW RESULTS CONNECT
1 Naming + positioning	Vynvori provisional; domain/trademark work continues separately.
2 Architecture	Core stack locked; benchmark selects exact multi-vendor models and router behavior.
3 Design Studio UX	Benchmark learnings shape concept/refinement UX and standard/premium model options.
4 Manufacturing + export	Printing constraints and sample outputs inform Bangladesh supplier tests and fulfillment quotes.
5 Unit economics + preorder	Measured AI cost becomes one input alongside blanks, printing, packaging, duties, delivery, marketing.
6 MVP build + rollout	Handoff contracts and cost controls become implementable tickets and launch acceptance tests.

THE NEXT MESSAGE TO SEND BACK WITH RESULTS

"Here is the completed Vynvori AI benchmark results folder. Evaluate the model leaderboard, actual billed costs, blind artwork scores, print suitability, and proposed credits. Recommend our standard/premium model stack and pricing. Then continue the main six-phase business plan from our next pending founder decisions."

Appendix D / Reference documentation and provenance

The following are starting sources, not permission to skip fresh verification. The coding agent must record the actual pages, dates and model endpoints checked during execution. Access baseline: 12 October 2026.

Google Gemini image generation - https://ai.google.dev/gemini-api/docs/image-generation
Google Gemini API pricing - https://ai.google.dev/gemini-api/docs/pricing
OpenAI image API guide - https://platform.openai.com/docs/guides/image-generation
OpenAI API pricing - https://openai.com/api/pricing/
fal hosted model and pricing list - https://fal.ai/pricing
Higgsfield developer quick start - https://open.higgsfield.ai/quick-start
Higgsfield developer API pricing - https://open.higgsfield.ai/pricing
Ideogram generation API docs - https://developer.ideogram.ai/v1/api-reference
Ideogram API pricing - https://about.ideogram.ai/api-pricing
Midjourney automation restrictions - https://docs.midjourney.com/hc/en-us/articles/32013696484109-Community-Guidelines
Vercel AI SDK image generation - https://ai-sdk.dev/docs/ai-sdk-core/image-generation
Vercel AI SDK provider registry - https://ai-sdk.dev/docs/reference/ai-sdk-core/provider-registry
Vercel Workflow SDK documentation - https://workflow-sdk.dev/
Cloudflare R2 docs - https://developers.cloudflare.com/r2/

Source-handling rule: Confirm model names, pricing, rate limits, output rights and SDK APIs directly. Model catalogs change rapidly; annotate any discrepancy between this blueprint and currently documented behavior.